Designing Role Vectors to Improve LLM Inference Behaviour

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Abstract

The influence of personas on Large Language Models (LLMs) has been widely studied, yet their direct impact on performance remains uncertain. This work explores a novel approach to guiding LLM behaviour through role vectors, an alternative to persona-based prompting. We construct 29 role vectors derived from model activations and evaluate their impact on benchmark performance across multiple domains. Our analysis investigates whether these vectors can effectively steer models toward domain-specific expertise. We measure two key interventions: (i) activation addition, which reinforces role-specific directions, and (ii) directional ablation, which removes them. Results on well-established benchmarks indicate that role vectors do, in fact, influence model behaviour, improving task performance in relevant domains while marginally affecting unrelated tasks. This, in turn, suggests that manipulating internal model representations has a greater impact on outcomes than persona-based prompting.

1 Introduction

The development of persona or role-based chatbots has gained significant attention in the AI and NLP community due to their potential impact on business and societal applications (Pataranutaporn et al., 2021). The extent to which different personas influence Large Language Models' (LLMs) performance on objective tasks remains unclear. Recent attempts investigate whether incorporating personas into system prompts enhances model performance on objective tasks and explores potential factors influencing these effects. (Zheng et al., 2024) conducted a large-scale analysis of the effect of personas in LLM prompting, examining the impact of domain alignment between personas and task-related questions, finding that persona-based prompting either has no effect or a slightly negative impact on model performance compared to a baseline setting.

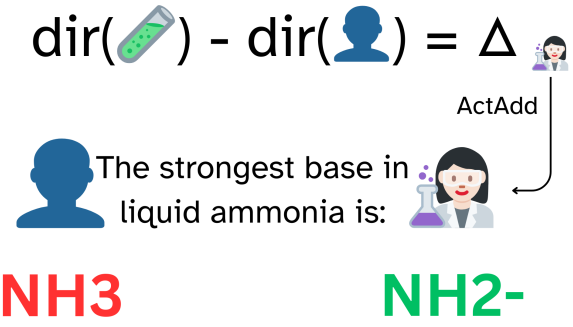

Figure 1: Illustrative example demonstrating how role vectors (e.g., chemist) can influence model outputs.

We aim to investigate whether modifying the model's internal mechanisms (Li et al., 2024), rather than a prompt-based approach, can lead to improved results. This forms the core objective of our current work, guided by the following research questions: **RQ1**: Can we identify specific latent role directions within the activation space, derived from the model's internal mechanisms, that, when leveraged, lead to improved performance on objective tasks? **RQ2**: Do the directions that enhance performance effectively impersonate the role of interest? **RQ3**: If we eliminate these directions in the models, do their performances suffer as a consequence?

1.1 Contribution

This work introduces a novel approach to guiding the behaviour of LLMs through role vectors, a structured method for embedding personas directly into model activations. Fig. 1 is an illustrative example showing how role vectors (in this example, a chemist-related vector) may impact LLM performance. Our key contributions are:

1. We develop 29 distinct role vectors for selected LLMs, each capturing domain-specific

knowledge and behavioural tendencies associated with each specialisation.

2. We investigate whether these vectors influence model behaviour on downstream benchmarks to determine whether explicit role-based directions in the activation space enhance model performance in domain-specific tasks.

3. Unlike traditional role-based prompting techniques, which show a limited or negative impact on performance, we show that role vector activation leads to measurable changes in model behaviour.

2 Preliminaries and State of the Art

Personas and Roles in LLMs. Decoder-only Transformers (Liu et al., 2018), commonly referred to as Language Models (LMs) and in their large form as LLMs, have gained increasing relevance over recent years. Prompting acts as a natural language interface facilitating human-AI interactions (Liu et al., 2023a). The effectiveness of LLMs is often dependent on prompt formulation (Lu et al., 2021); for instance, including the phrase "Let's think step by step" can enhance model performance across a variety of tasks and queries (Kojima et al., 2022). Prior research explored the impact of in-context impersonation tasks (Salewski et al., 2023), demonstrating that incorporating socio-demographic details can be advantageous for subjective NLP tasks in zero-shot scenarios. Other studies highlight the potential biases and constraints associated with persona-based and socio-demographic-driven prompting (Sun et al., 2023; Hu and Collier, 2024; Beck et al., 2024). (Zheng et al., 2024) shows that adding personas does not consistently enhance performance on objective tasks and, in some cases, may even degrade it.

In this work, we explore whether alternative approaches associated with Representation Engineering and Mechanistic Interpretability can successfully inject roles into LLMs and yield distinct performance outcomes compared to conventional persona-based or socio-demographic prompting strategies.

Mechanistic Interpretability and Directions. Pioneering research by Anthropic and other scholars (Elhage et al., 2022; Bolukbasi et al., 2016; Hernandez and Andreas, 2021) has demonstrated that neural networks encode input attributes as specific directions within the activation space. It is well established that introducing feature vectors into the residual stream of LMs can influence network behaviour (Li et al., 2024; Turner et al., 2023; Arditi et al., 2024), though the precise mechanisms and optimal intervention points remain an area of active investigation (Jorgensen et al., 2023; von Rütte et al., 2024).

The study by (Scalena et al., 2024) demonstrates that steering can effectively induce behavioural modifications in LMs, such as facilitating language switching while providing insights into how specific properties influence model behaviour during text generation. (Zhao et al., 2024) introduce SPARE, a training-free representation engineering technique that leverages pre-trained Sparse Autoencoders (SAEs) (Cunningham et al., 2023) to enable controlled selection of the model's knowledge during inference. Similarly, (Li et al., 2024) proposes ITI, a method to enhance the truthfulness of LM outputs by employing supervised learning to identify latent vectors associated with factual responses and adjusting model activations during inference. This approach demonstrates improved performance on the TruthfulQA (Lin et al., 2021) and MMLU (Hendrycks et al., 2020) benchmarks relative to the baseline.

The results obtained from these works motivate us to explore this class of methodologies to investigate alternative methods for role injection beyond prompting.

3 Generating and Evaluating Role Vectors

The methodology is composed of three main components: (i) personas selection and prompt dataset generation, (ii) selection of relevant directions and (iii) evaluation methods.

3.1 Personas Selection and Dataset Generation

To systematically assess the models' knowledge and reasoning capabilities across various domains, our study adopts a role-based evaluation framework inspired by (Zheng et al., 2024), inheriting 29 distinct roles $R = \{r_1, r_2, \ldots, r_{29}\}$, each $r \in R$ associated with a unique professional or academic specialisation (see Tab. 2). Roles not corresponding to an occupation or not associated with any PersonaHub personas were excluded.

The prompt roles dataset used to identify specific role directions is extracted using the correspond-

ing personas for each role from PersonaHub (Ge et al., 2024). These personas are highly specialised and situated in realistic settings and represent various contextualised scenarios, such as *"A pharmaceutical chemist who analyses the chemical properties of medical devices"*. First, we perform strict string matching to identify personas that explicitly contain the role name. That is, for each role r, we obtain $P(r) = \{p \in \text{PersonaHub} \mid \text{string-match}(p, r)\}$ where $\text{string-match}(p, r)$ indicates that the persona p explicitly contains the role name r. Then, a sampling process is applied to select relevant personas randomly. Each role can have one or multiple personas, ranging from a minimum of 1 to a maximum of 6948 (881 on average).

The selected personas are then used to generate a synthetic dataset for each role, following the methodology employed by Alpaca (Taori et al., 2023) to create its dataset.

We define a set of tasks $T = \{\text{write}, \text{explain}, \text{design}, \text{what is}, \text{how to}, \ldots\}$, analogous to those used in Alpaca (Taori et al., 2023). We generate a set of prompts for each role $r \in R$. Let $D_r = \{x_{r,1}, x_{r,2}, \ldots, x_{r,128}\}$ be the collection of 128 prompt examples for role r. For each prompt $x_{r,i}$, a task t is randomly sampled from T, and a persona p is randomly sampled from $P(r)$. Then, the prompt is generated by providing the template (see Fig. 2) to the Claude 3.5 Haiku model (Anthropic, 2024b) with the selected task t and persona p.

Generating Persona-Specific Tasks

Generate a *{task_type}* prompt that this persona would likely ask: Persona: *{persona}*.

Rules: (i) The prompt should start with "*{task_type}*". (ii) Keep it specific and under 15 words. (iii) Make it relevant to the persona's background/interests. (iv) Your output must start with "User prompt:".

Examples based on task types:

- *Describe*: "Describe the key features of a successful marketing campaign."
- *Explain*: "Explain the process of setting up a home network."
- *Design*: "Design a logo for a sustainable fashion brand."
- *What is*: "What is the difference between UI and UX design?"
- *How to*: "How to optimise a website for mobile devices?"

Figure 2: Prompt template for generating persona-specific tasks.

The complete roles tasks dataset is given by $\mathcal{D}_{\text{roles}} = \bigcup_{r \in R} D_r$. We incorporate $\mathcal{D}_{\text{base}}$, an additional set of 128 examples sourced from the original Alpaca dataset, consisting of general instruction-following prompts. This provides a broad reference point, enabling the contrastive computation of direction for each role using the corresponding D_r.

3.2 Selection of Role Directions

Our evaluation of steering effects utilizes the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2020), adhering to the sampling and splitting methodology described in (Zheng et al., 2024) for a total of 2457 questions. We define the set of categories as $C = \{$*Natural Science, Economics, EECS (Electrical Engineering and Computer Sciences), Law, Math, Medicine, Politics, Psychology*$\}$.

For each $c \in \mathcal{C}$, let D_c denote the set of questions corresponding to category c. The overall test dataset is then defined as $\mathcal{D}_{\text{test}} = \bigcup_{c \in \mathcal{C}} D_c$. Tab. 1 shows the distribution of questions. While many of these roles correspond directly to established MMLU categories, some exhibit only partial alignment. For example, the role of a dentist does not perfectly fit within the "medicine" category. However, we expect that individuals or models adopting the role of a dentist should demonstrate domain-specific knowledge that exceeds that of the general population or those assuming unrelated roles.

Category	# Questions
Natural Science	590
Economics	492
EECS	247
Law	200
Math	287
Medicine	241
Politics	200
Psychology	200
Total	2457

Table 1: Number of questions per category in $\mathcal{D}_{\text{test}}$., taken from (Zheng et al., 2024)

To identify the direction in the model's residual stream activations corresponding to each role, we use a technique known as *difference-in-means* (Belrose, 2023): we compute the difference between the model's average activations when performing inference on the role-specific dataset $D_r \in \mathcal{D}_{\text{roles}}$ and generic queries from $\mathcal{D}_{\text{base}}$.

Following the notation from (Arditi et al., 2024), for each role $r \in R$, layer $l \in [L]$, and post-instruction token position $i \in I$, we compute the mean activation $\mu_{i,r}^{(l)}$ for role-specific prompts in D_r and $\nu_i^{(l)}$ for generic prompts in $\mathcal{D}_{\text{base}}$: